Build a prediction one coordinate at a time

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Reference frame / ES: sistema de referencia / PT: referencial

Keep the origin and axes fixed.

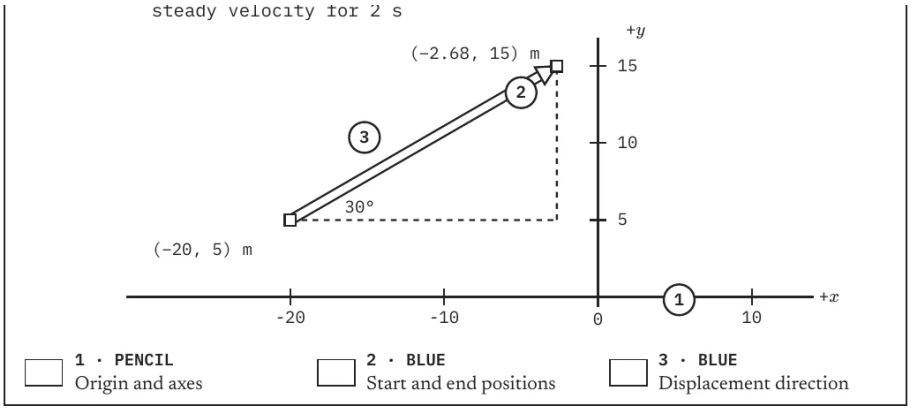
Also on your note page: Components, Units, Limit.

B READ THE RELATIONSHIP

Advance x and y independently using vx and vy in the same frame.

As you read, find: origin · direction · assumption.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A probe starts at (−20, 5) m and moves at 10 m/s , 30° from + x toward + y , for 2 s . Find its new coordinates.

Given x 0 = −20 m ; y 0 = 5 m v = 10 m/s ; θ = 30° t = 2 s ; steady v

Rule Resolve v : v x = 8.66 m/s v y = 5 m/s

Work x f = −20 + 8.66(2) y f = 5 + 5(2)

Answer (−2.68, 15) m Final x is still negative.

E SAVE FOR THE FINAL MEETING

For fictional XJ, name the frame, units and motion assumptions. Fixed Earth and steady velocity are classroom simplifications.

Read the physics: Reference frame

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Name a reference point, mark the origin and choose axes. Record the starting position and the time. Include a unit with each value.

02 HOW IT WORKS

If velocity is steady, add velocity times elapsed time to the starting coordinate. In two dimensions, do that once for x and once for y. Use the matching component each time.

03 WHAT TO CHECK

Check whether the final position is on the expected side of zero. Say which motion assumption could fail when a new sighting arrives.

04 TRY THE EXAMPLE FROM YOUR NOTES

A probe starts at (−20, 5) m and moves at 10 m/s , 30° from + x toward + y , for 2 s . Find its new coordinates. (−2.68, 15) m Final x is still negative.

Model check: Advance x and y independently using vx and vy in the same frame.

05 CONNECT TO XJ

For fictional XJ, name the frame, units and motion assumptions. Fixed Earth and steady velocity are classroom simplifications.

SAY IT BEFORE YOU WRITE

Would moving zero change the motion or only its coordinates? From ___ as zero, I predict ___ at time ___. I assumed ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Reference frame

ES: sistema de referencia / PT: referencial

Meaning in my words	How can you tell?
How would you explain reference frame to a friend? In my words, this means ___.	You report a cart at +5 m. What does a partner need to locate it? I can tell because ___.
Example or sketch	**Non-example and why**
How could a seated passenger be at rest in one frame and moving in another? My example shows ___.	Two people report different velocities. Does that alone prove one measured incorrectly? I would change ___ because ___.

Components

ES: componentes / PT: componentes

Meaning in my words	How can you tell?
What are the components of an arrow? In my words, this means ___.	If east and north are positive, which component is negative for a southeast trip? I can tell because ___.
Example or sketch	**Non-example and why**
What two arrows could you use to build a displacement of 3 m east and 4 m north? My example shows ___.	A classmate counts the diagonal and both parts as three separate trips. What is counted twice? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Units

ES: unidades / PT: unidades

Meaning in my words	How can you tell?
What do units tell you about a number? In my words, this means ___.	Before adding 2 m and 30 cm, what should you do? I can tell because ___.
Example or sketch	**Non-example and why**
How would you add those two lengths in metres? My example shows ___.	A partner adds the numbers and gets 32 m. What went wrong? I would change ___ because ___.

Limit

ES: límite / PT: limite

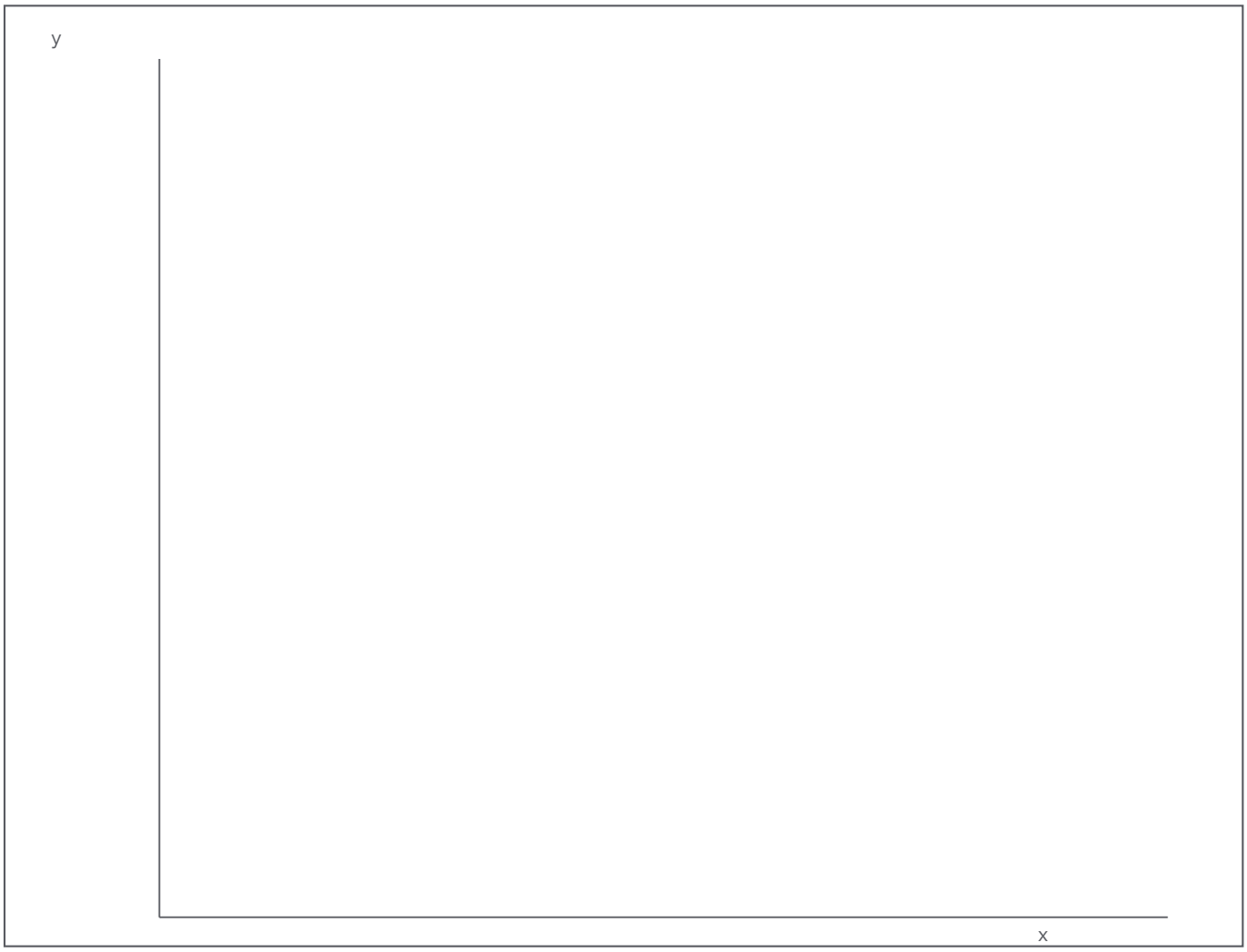
Meaning in my words	How can you tell?
How would you explain limit to a friend? In my words, this means ___.	Which change would make a steady-velocity forecast unreliable? I can tell because ___.
Example or sketch	**Non-example and why**
What limit would you write next to a one-minute straight-line prediction? My example shows ___.	A partner uses the one-minute model for a year. What evidence would they need first? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the reference frame and one sideways component. Box your time unit and one model assumption.

Labels you can use: origin · direction · assumption.



Tell a partner

From ___ as zero, I predict ___ at time ___. I assumed ___.

On assignment: Reference frame

SECTION E / BACK / FINAL MEETING ONLY Name _____

Two observatories file the same sighting: “5 km right of station A” and “2 km left of station B.” B is 7 km right of A.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Survey engineer**

Put both stations on one axis. Give dispatch one position in A’s frame, then list what else a future prediction needs.

☐ **Map broadcaster**

Make an on-air locator map. Convince the producer whether the two reports disagree by showing both zeros.

Word bank: origin · direction · assumption.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”

Editor’s check: Name a bound that should travel with your central forecast.